

Exercise 2.37 :

1) Let h be in $\mathcal{L}^0(E, \mathbb{R}_+)$.

By transfer, $\mathbb{E}(h(X)) = \int_E h(x) \mathbb{P}_X(dx) = \int_E h(x) \mathbb{P}_Y(dx) = \mathbb{E}(h(Y))$.

2) With $h = \mathbb{1}_B$ and the previous question, we obtain the result.

Exercise 2.38 :**E.2.1 :**

- Let $B \in \mathcal{B}(\mathbb{R})$,
 $\mathbb{P}(X \in B) = \frac{\lambda([a,b] \cap B)}{\lambda([a,b])} = \frac{1}{b-a} \int_{\mathbb{R}} \mathbb{1}_{[a,b] \cap B}(x) dx = \int_B \frac{1}{b-a} \mathbb{1}_{[a,b]}(x) dx$.
Hence, $\rho_X = \frac{1}{b-a} \mathbb{1}_{[a,b]}$.
- Let t be in $\mathbb{R}$.
 $F_X(t) = \mathbb{P}(X \leq t) = \frac{\lambda([a,b] \cap]-\infty, t])}{b-a}$.
If $t \leq a$, $[a,b] \cap]-\infty, t] = \emptyset$ and $F_X(t) = 0$.
If $a \leq t \leq b$, $[a,b] \cap]-\infty, t] = [a, t]$ and $F_X(t) = \frac{t-a}{b-a}$.
If $t \geq b$, $[a,b] \cap]-\infty, t] = [a, b]$ and $F_X(t) = 1$.
- By transfer, $\mathbb{E}(h(X)) = \int_{\mathbb{R}} h(x) \rho(x) dx = \frac{1}{b-a} \int_{\mathbb{R}} h(x) \mathbb{1}_{[a,b]} dx = \frac{1}{b-a} \int_a^b h(x) dx$.
- With $h : x \rightarrow x^2$, we have $\mathbb{E}(X^2) < +\infty$.
Then, with $h : x \rightarrow x$, $\mathbb{E}(X) = \frac{1}{b-a} \int_a^b x dx = \frac{b^2-a^2}{2(b-a)} = \frac{b+a}{2}$.
We also have that $V(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$ and because $\mathbb{E}(X^2) = \frac{1}{b-a} \int_a^b x^2 dx = \frac{b^3-a^3}{3(b-a)} = \frac{a^2+ab+b^2}{3}$,
 $V(X) = \frac{a^2+ab+b^2}{3} - \frac{a^2+2ab+b^2}{4} = \frac{(b-a)^2}{12}$.

E.2.2 :

- Let $B \in \mathcal{B}(\mathbb{R})$,
 $\mathbb{P}(X \in B) = \frac{1}{\sigma\sqrt{2\pi}} \int_B e^{-\frac{(x-m)^2}{2\sigma^2}} dx$. By identification, we have the result.
- Let t be in $\mathbb{R}$. Now we have $B =]-\infty, t]$
 $F_X(t) = \mathbb{P}(X \leq t) = \frac{1}{\sigma\sqrt{2\pi}} \int_{]-\infty, t]} e^{-\frac{(x-m)^2}{2\sigma^2}} dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^t e^{-\frac{(x-m)^2}{2\sigma^2}} dx$.
- By transfer, $\mathbb{E}(h(X)) = \int_{\mathbb{R}} h(x) \rho(x) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{\mathbb{R}} h(x) e^{-\frac{(x-m)^2}{2\sigma^2}} dx$.
- With $h : x \rightarrow x^2$, we have $\mathbb{E}(X^2) < +\infty$.
Then, with $h : x \rightarrow x$, $\mathbb{E}(X) = \frac{1}{\sigma\sqrt{2\pi}} \int_{\mathbb{R}} x e^{-\frac{(x-m)^2}{2\sigma^2}} dx$.
With the change of variable $\phi : \begin{cases} \mathbb{R} \longrightarrow \mathbb{R} \\ x \longmapsto m + \sigma x \end{cases}$ and $\phi^{-1} : \begin{cases} \mathbb{R} \longrightarrow \mathbb{R} \\ y \longmapsto \frac{y-m}{\sigma} \end{cases}$,

$dx = \sigma dy$, we have

$$\mathbb{E}(X) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} (m + \sigma y) e^{-y^2/2} dy = \frac{m}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-y^2/2} dy + \frac{\sigma}{\sqrt{2\pi}} \int_{\mathbb{R}} y e^{-y^2/2} dy.$$

But we know that

$$\int_{\mathbb{R}} e^{-y^2/2} dy = \sqrt{2\pi}$$

and because $y e^{-y^2/2}$ is an odd function, $\int_{\mathbb{R}} y e^{-y^2/2} dy = 0$.

Hence,

$$\mathbb{E}(X) = m.$$

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Then, we use the following formula :

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2.$$

We have

$$\mathbb{E}(X^2) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} (m + \sigma y)^2 e^{-y^2/2} dy =$$

$$\frac{m^2}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-y^2/2} dy + \frac{2m\sigma}{\sqrt{2\pi}} \int_{\mathbb{R}} y e^{-y^2/2} dy + \frac{\sigma^2}{\sqrt{2\pi}} \int_{\mathbb{R}} y^2 e^{-y^2/2} dy.$$

We know that

$$\int_{\mathbb{R}} e^{-y^2/2} dy = \sqrt{2\pi}$$

and

$$\int_{\mathbb{R}} y e^{-y^2/2} dy = 0.$$

We have to generalize the integration by parts. Thus we need to shorten the boundaries of the integrals. Let us denote

$$I_n = \int_{-n}^n y^2 e^{-y^2/2} dy.$$

We integrate by parts :

$$u = y, \quad dv = y e^{-y^2/2} dy,$$

thus

$$du = dy, \quad v = -e^{-y^2/2}$$

Hence

$$I_n = \left[-ye^{-y^2/2} \right]_{-n}^n + \int_{-n}^n e^{-y^2/2} dy.$$

First

$$\left[-ye^{-y^2/2} \right]_{-n}^n = -ne^{-n^2/2} - ne^{-n^2/2} = -2ne^{-n^2/2}.$$

Then

$$I_n = -2ne^{-n^2/2} + \int_{-n}^n e^{-y^2/2} dy.$$

When n goes to infinity

$$\lim_{n \rightarrow \infty} 2ne^{-n^2/2} = 0,$$

and

$$\lim_{n \rightarrow \infty} \int_{-n}^n e^{-y^2/2} dy = \int_{\mathbb{R}} e^{-y^2/2} dy = \sqrt{2\pi}.$$

Finally

$$\int_{\mathbb{R}} y^2 e^{-y^2/2} dy = \lim_{n \rightarrow \infty} I_n = \sqrt{2\pi}.$$

To conclude,

$$\mathbb{E}(X^2) = m^2 + \sigma^2.$$

and

$$\text{Var}(X) = (m^2 + \sigma^2) - m^2 = \sigma^2.$$

- We need to prove that for $h \in \mathcal{L}^0(\mathbb{R}, \mathbb{R}_+)$,

$$\mathbb{E}(h(X + X')) = \frac{1}{\sqrt{\sigma^2 + \sigma'^2} \sqrt{2\pi}} \int_{\mathbb{R}} h(x) e^{-\frac{(x-m-m')^2}{2(\sigma^2 + \sigma'^2)}} dx.$$

First, assume that $m = m' = 0$ and $\sigma = \sigma' = 1$.

Hence $X \sim \mathcal{N}(0, 1)$, $X' \sim \mathcal{N}(0, 1)$.

Let us denote $G = \sigma X + \sigma' X'$. The density of G is

$$\rho_2(y_1, y_2) = \frac{1}{2\pi} e^{-(y_1^2 + y_2^2)/2}.$$

For $h \in \mathcal{L}^0(\mathbb{R}, \mathbb{R}_+)$,

$$\mathbb{E}[h(G)] = \int_{\mathbb{R}^2} h(\sigma_1 y_1 + \sigma_2 y_2) \rho_2(y_1, y_2) dy_1 dy_2.$$

We use the following change of variables :

$$u = \sigma_1 y_1 + \sigma_2 y_2, \quad v = -\sigma_2 y_1 + \sigma_1 y_2.$$

$$J_\theta = \begin{pmatrix} \sigma & \sigma' \\ -\sigma' & \sigma \end{pmatrix},$$

$$\det J_\theta = \sigma^2 + \sigma'^2 =: \mu.$$

We verify that

$$y_1^2 + y_2^2 = \frac{u^2 + v^2}{\mu^2}, \quad dy_1 dy_2 = \frac{1}{\mu^2} du dv.$$

Thus

$$\mathbb{E}[h(G)] = \int_{\mathbb{R}^2} h(u) \frac{1}{2\pi} \exp\left(-\frac{u^2 + v^2}{2\mu^2}\right) \frac{1}{\mu^2} du dv.$$

By Fubini,

$$\mathbb{E}[h(G)] = \int_{\mathbb{R}} h(u) \left(\frac{1}{\mu\sqrt{2\pi}} e^{-u^2/(2\mu^2)} \right) \left(\int_{\mathbb{R}} \frac{1}{\mu\sqrt{2\pi}} e^{-v^2/(2\mu^2)} dv \right) du.$$

However, due to

$$\int_{\mathbb{R}} \frac{1}{\mu\sqrt{2\pi}} e^{-v^2/(2\mu^2)} dv = 1,$$

we have

$$\mathbb{E}[h(G)] = \int_{\mathbb{R}} h(u) \frac{1}{\mu\sqrt{2\pi}} e^{-u^2/(2\mu^2)} du.$$

Then $G \sim \mathcal{N}(0, \mu^2)$.

For the general case, let us denote $Y = m + \sigma X$, $Y' = m' + \sigma' X'$.

Then $Y + Y' = (m + m') + G$, and

$$\mathbb{E}(h(Y + Y')) = \mathbb{E}(h(m + m' + G)) = \int_{\mathbb{R}} h(m + m' + u) \frac{1}{\mu\sqrt{2\pi}} e^{-u^2/(2\mu^2)} du =$$

$$\int_{\mathbb{R}} h(v) \frac{1}{\mu\sqrt{2\pi}} e^{-(v-m-m')^2/(2\mu^2)} dv$$

by a change of variables ($v = m + u$).

Finally

$$Y + Y' \sim \mathcal{N}(m + m', \sigma^2 + \sigma'^2).$$

E.2.2 :

- Let $B \in \mathcal{B}(\mathbb{R})$,
 $\mathbb{P}(X \in B) = \int_{B \cap \mathbb{R}_+} \alpha e^{-\alpha x} dx = \int_B \alpha e^{-\alpha x} \mathbb{1}_{x \geq 0}(x) dx$. By identification, we have the result.

- Let t be in $\mathbb{R}$. Now we have $B =]-\infty, t]$
 $F_X(t) = \mathbb{P}(X \leq t) = \int_{]-\infty, t]} \alpha e^{-\alpha x} \mathbb{1}_{x \geq 0}(x) dx = \int_0^t \alpha e^{-\alpha x} dx$.
If $t < 0$, $F_X(t) = 0$.
If $t \geq 0$, $F_X(t) = 1 - e^{-\alpha t}$.
- By transfer, $\mathbb{E}(h(X)) = \int_{\mathbb{R}} h(x) \rho(x) dx = \int_{\mathbb{R}} h(x) \alpha e^{-\alpha x} \mathbb{1}_{x \geq 0}(x) dx$
 $= \alpha \int_0^{+\infty} h(x) e^{-\alpha x} dx$.
- With $h : x \rightarrow x^2$, we have $\mathbb{E}(X^2) < +\infty$.
Then, with $h : x \rightarrow x$, $\mathbb{E}(X) = \alpha \int_{\mathbb{R}_+} x e^{-\alpha x} dx$.
Let us denote for $n > 0$,

$$I_n := \alpha \int_0^n x e^{-\alpha x} dx.$$

We have

$$\mathbb{E}(X) = \lim_{n \rightarrow +\infty} I_n.$$

With an integration by parts, such that

$$u = x, \quad dv = e^{-\alpha x} dx,$$

and

$$du = dx, \quad v = -\frac{1}{\alpha} e^{-\alpha x},$$

We have

$$\int_0^n x e^{-\alpha x} dx = \left[-\frac{x}{\alpha} e^{-\alpha x} \right]_0^n + \frac{1}{\alpha} \int_0^n e^{-\alpha x} dx.$$

First,

$$\left[-\frac{x}{\alpha} e^{-\alpha x} \right]_0^n = -\frac{n}{\alpha} e^{-\alpha n} - 0,$$

and

$$\int_0^n e^{-\alpha x} dx = \left[-\frac{1}{\alpha} e^{-\alpha x} \right]_0^n = -\frac{1}{\alpha} e^{-\alpha n} + \frac{1}{\alpha}.$$

Hence

$$I_n = \alpha \int_0^n x e^{-\alpha x} dx = \alpha \left(-\frac{n}{\alpha} e^{-\alpha n} - \frac{1}{\alpha^2} e^{-\alpha n} + \frac{1}{\alpha^2} \right) = -n e^{-\alpha n} - \frac{1}{\alpha} e^{-\alpha n} + \frac{1}{\alpha}.$$

When n goes to the infinity,

$$ne^{-\alpha n} \longrightarrow 0, \quad e^{-\alpha n} \longrightarrow 0,$$

and

$$\lim_{n \rightarrow +\infty} I_n = \mathbb{E}(X) = \alpha \int_0^{+\infty} xe^{-\alpha x} dx = \frac{1}{\alpha}.$$

Then, we use the following formula :

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2.$$

We have

$$\mathbb{E}(X^2) = \alpha \int_{\mathbb{R}_+} x^2 e^{-\alpha x} dx =$$

Similarly, let us denote for $n > 0$,

$$J_n := \int_0^n x^2 \alpha e^{-\alpha x} dx, \quad \mathbb{E}(X^2) = \lim_{n \rightarrow +\infty} J_n.$$

With an integration by parts, such that

$$u = x^2, \quad dv = \alpha e^{-\alpha x} dx \quad \Rightarrow \quad du = 2x dx, \quad v = -e^{-\alpha x}.$$

Hence,

$$J_n = \left[-x^2 e^{-\alpha x} \right]_0^n + \int_0^n 2x e^{-\alpha x} dx = -n^2 e^{-\alpha n} + 2 \int_0^n x e^{-\alpha x} dx.$$

We already have :

$$\int_0^n x e^{-\alpha x} dx = -\frac{n}{\alpha} e^{-\alpha n} - \frac{1}{\alpha^2} e^{-\alpha n} + \frac{1}{\alpha^2}.$$

Finally

$$J_n = -n^2 e^{-\alpha n} - \frac{2n}{\alpha} e^{-\alpha n} - \frac{2}{\alpha^2} e^{-\alpha n} + \frac{2}{\alpha^2}.$$

When n goes to the infinity,

$$n^2 e^{-\alpha n} \rightarrow 0, \quad n e^{-\alpha n} \rightarrow 0, \quad e^{-\alpha n} \rightarrow 0,$$

And

$$\mathbb{E}(X^2) = \lim_{n \rightarrow +\infty} J_n = \frac{2}{\alpha^2}.$$

Then

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{2}{\alpha^2} - \left(\frac{1}{\alpha}\right)^2 = \frac{2}{\alpha^2} - \frac{1}{\alpha^2} = \frac{1}{\alpha^2}.$$